

ClarkNet-HTTP Server Analysis

File analysis results

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1.Overview :

About ClarkNet:

- **ClarkNet** was one of the first **independent commercial Internet Service Providers** in the United States, based in Maryland.
- The dataset originates from the **public web server** of ClarkNet, which hosts user and business web pages.
- It served content via **HTTP/1.0**, mostly static files such as .html, .gif, and .jpg.

Log Dataset Details:

- Analyzed file: **clarknet_access_log_Sep4**
- Total requests: **1,673,750**
- Date range covered: **September 4–10, 1995**
- Format: **Apache Common Log Format**
- Unique IPs: **94,772**
- Total 404 errors: **35,542**

Server Stack :

- **Operating System:** Unix-based
- **Web Server:** Apache HTTPD **or** NCSA HTTPD
- **Content Type:** Fully static; (no AJAX or JavaScript-based interactivity)

2.Summary of Script Output Analysis

Request Statistics:

- Total number of requests: 1,673,750
GET requests: 1,669,515
 - POST requests: 2,538
 - Unique IP addresses: 94,772
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Requests by IP:

- Most active IP: piweba5y.prodigy.com with 7,239 requests
 - Top 10 most active IPs are primarily from AOL proxies and IBM networks
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Failed Requests :

- Total failed requests: 37,975
 - Failure rate: 2.27%
 - Days with highest failure counts:
 - September 8: 6,485
 - September 7: 6,469
 - September 6: 6,219
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Daily Activity:

- Busiest day: September 8 with 280,342 requests
 - Least active day: September 10 with 192,263 requests
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Hourly Distribution:

- Peak activity hours: Between 12:00 PM and 3:00 PM
 - Lowest activity: Early morning hours between 03:00 and 06:00 AM
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HTTP Status Code Breakdown:

- 200 OK: 1,485,020
 - 304 Not Modified: 134,782
 - 404 Not Found: 35,542 <- relatively high
 - 500 Internal Server Error: 1,303
 - 403 Forbidden: 958
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Most Requested Pages:

- Product catalog GIF images:
 - </pub/atomicbk/catalog/logo2.gif> -> over 20,000 times
 - Popular HTML pages:
 - </pub/journalism/awesome.html>
 - </pub/job/vk/vendela.html>
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Top 404 Error Pages:

- </pub/atomicbk/images/earlyero.gif>
 - </pub/atomicbk/images/nudistbk.gif>
 - </pub/mjoneill/links.html>
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Most Requested File Types:

- .gif: 839,465
- .jpg: 421,556
- .html: 276,342
- Others like .txt, .wav, .zip, .pl, .cgi were far less frequent

3. Analysis Insights & Observations

Request Trend Analysis

- **Peak Day:** September 8, 1995 — with 280,342 requests.
 - **Lowest Activity:** September 10, 1995 — with 192,263 requests.
 - **Overall Pattern:** There is a steady increase in request volume from September 4 to 8, followed by a noticeable decline on September 9 and 10. This pattern indicates typical weekday traffic peaks and weekend drops.
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Failure Requests by Hour

- **The highest failure activity occurred during:**
 - 16:00 → 2,359 failed requests
 - 14:00 → 2,220 failed requests
 - 17:00 → 2,099 failed requests
 - These hours correspond to high traffic periods, suggesting that the failures could be due to server load or targeted probing attempts.
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GET vs POST Breakdown per IP (Top 10)

- **Example:**
 - **piweba5y.prodigy.com: 7,219 GET and 20 POST requests**
 - **www-e5.proxy.aol.com: 5,122 GET and 1 POST**
- **Observation:** The majority of top IPs use only GET, which is consistent with browsing or automated scanning. A few IPs also show POST activity that may require deeper inspection.

4.Detailed Security notes from log analysis with suggested improvements

1. IPs Sending Over 100 Requests per Hour

Observtion :

- These are likely automated bots, vulnerability scanners, **or light DoS attempts.**

Risk :

- Can lead to resource exhaustion or prep for **brute-force attacks.**

Action needed :

- Implement **rate-limiting**, monitor suspicious IPs, and use tools like **fail2ban** to auto-block them.
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2. High Volume of 404 Not Found Errors

Observtion :

- Many clients are attempting to access **nonexistent files.**

Ex :

- **/admin, /backup.zip, or typo variations like /html.**

Action needed :

- Monitor recurring 404 patterns.
- block IPs that trigger too many in a short time.

3. Heavy Traffic on Static Media (.gif, .jpg)

Observtion :

- Indicates automated scripts may be scraping site content or causing bandwidth.

Risk :

- Can slow down the site or lead to extra hosting costs.

Action needed :

- Use **anti-scraping headers**, **CDN protections**, or tools like **Cloudflare** with **bot mitigation enabled**.
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4. Repeated Requests for the Same Files from Multiple IPs

Observtion :

- This could signal distributed probing, where multiple IPs are scripted to try the same URLs to avoid detection.

Action needed :

- Use log correlation to track this pattern.
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5. Consistent Traffic Spikes During Specific Hours

Observtion :

- Traffic peaks between **12 PM and 3 PM** suggest these hours are the busiest — which also makes them prime time **for attacks to blend in.**

Action needed :

- Strengthen server performance and enable protective measures during peak hours, like **WAF or load balancing.**
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6. Unusual POST Activity

Observtion :

- While POST requests are **low in number**, any unexpected POST traffic should be closely analyzed.

Risk :

- **POST** is often used in attacks like SQL Injection, file upload abuse, or credential stuffing.

Action needed :

- **Log POST** data paths and parameters.
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7. Suspicious IPs Involved in Multiple Threat Patterns

Observtion :

- Some IPs appear in multiple suspicious contexts “excessive **GETs, POSTs, and 404s** “ indicating they are actively probing the system.

Action needed :

- **Add such IPs** to blocklists or greylists, and flag them for deeper analysis.
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8. Presence of HTTP 500 Errors

Observtion :

- Internal server errors may indicate backend instability or trying to crash or exploit the server.

Action needed :

- **Correlate 500 errors with their IP origin and request path.**
 - review backend logs.
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9. Targeted 404 Errors for Specific Filenames

Observation:

- Filenames like nudistbk.gif or links.html suggest bots are looking for sensitive or private content, not just random file guessing.

Risk:

- Indicates targeted probing, potentially based on known file structures or previously leaked data.

Action needed:

- Track and alert on specific keyword-based URL attempts.
 - consider using honey URLs to catch attackers in the act.
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10. Unusual File Types Being Requested

Observation:

- Requests for .cgi, .pl, .zip, or .bak files often indicate attempts to:
 - Execute old vulnerable scripts.
 - Download backup files left on the server.

Risk:

- May lead to information disclosure or remote code execution.

Action needed:

- Restrict public access to these file types using .htaccess or web server configs.

5.Conclusion

The analyzed log file from the ClarkNet web server provides valuable insights into real-world web traffic behavior from 1995. It reveals common patterns such as high volumes of GET requests, a significant number of 404 errors, and potential signs of automated scanning or denial-of-service attempts. Several IPs showed suspicious activity, making it essential to implement proper security measures like request rate-limiting, 404 monitoring, and protection against content scraping. Overall, the log highlights the importance of regular log analysis for detecting abnormal behavior and strengthening server security.

Thank You
Ahmed Mohammed